Source-Scoped Evidence, Path-Relevant Truth, and Weakness-Guided Cognitive Reasoning

A modular foundation for NARS, PLN, enactive generalization, and resource-bounded agents

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Abstract

Reasoning systems often agree on a numerical readout while disagreeing about what that number means, which evidence it summarizes, and which operations it licenses. This paper develops an information-first architecture in which an evidential object retains its source scope, derivation path, and multiplicity; truth values, probabilities, confidence scores, and Boolean verdicts are derived observations. The architecture clarifies four related bodies of work. Pei Wang's NARS motivates path- and scope-relative truth under insufficient knowledge and resources. Probabilistic Logic Networks (PLN) and constructible duality contribute positive/negative evidence geometry without forcing NARS into a globally coherent probability model. Michael Timothy Bennett's weakness orders admissible hypotheses by semantic freedom rather than description length, while Ben Goertzel's quantale-valued weakness makes relational composition explicit. Eray Özkural's task-process models expose the distributional assumptions under which weakness predicts generalization, and Patrick Hammer's ONA work supplies a concrete bounded control layer and a clean observation-aliasing problem.

The common foundation is deliberately weak: finite source geometry and a constrained preorder of admissible candidates. NARS and PLN remain sibling theories over this substrate, connected only by explicit bridge theorems. Positive and negative controls show why source identity cannot be replaced by source count, why equal intervals need not imply equal semantics, why Bennett's Razor and shortest-description criteria do not subsume one another, and why nonuniform task processes require additional premises. The result is a reusable primer for proof-relevant cognitive languages and their formal development.

Contents

1 The information-first method

The central methodological rule is simple.

Principle 1.1 (Informative object before quotient)**.** *Build the object that retains distinctions required by later reasoning. A lossy observable may be derived from it, but the observable should not be installed as the foundation unless a factorization theorem proves that no required distinction is lost.*

For evidential reasoning, the informative object may include

$$(\text{claim},\ \text{derivation path},\ \text{source scope},\ \text{positive evidence},\ \text{negative evidence},\ \text{receipt}).$$

A frequency, confidence, interval, probability, Boolean status, or scalar cost is then a readout of this object. Different readouts answer different questions. Equal readouts do not identify the objects that produced them.

This distinction matters whenever evidence can be duplicated, shared, or reached through multiple derivations. Suppose two proofs conclude the same formula. They may be:

(a) independent proofs from disjoint sources;

(b) two views of the same source;

(c) different rule paths that partially overlap;

(d) distinct occurrences of an otherwise identical derivation.

Counting all four alike double-counts evidence. Deduplicating all four alike erases genuine multiplicity. The remedy is not a better scalar; it is to retain enough identity to decide which quotient is appropriate.

2 The weakest common evidence substrate

Let S be a type of source occurrences. A packet type P is *source-scoped* when it carries a finite source map

$$\mathsf{scope}_P : P \longrightarrow \mathsf{Finset}(S).$$

No truth-value representation, probability interpretation, or merge policy is part of this definition.

Definition 2.1 (Source independence)**.** *For source-scoped packets $p : P$ and $q : Q$ over the same source vocabulary, possibly with $P \neq Q$, define*

$$p \#_S q \quad \iff \quad \mathsf{scope}_P(p) \cap \mathsf{scope}_Q(q) = \varnothing.$$

The heterogeneous form is essential. A NARS belief and a PLN evidence packet need not share a carrier type in order to reveal that they cite the same observation. Their commonality is provenance, not identity of logic.

Proposition 2.2 (Elementary source laws)**.** *Source independence is symmetric. A packet is independent of itself exactly when its scope is empty. If a source belongs to both scopes, the packets are not independent.*

Example 2.3 (Positive control)**.** *A direct observation cites source s_0; a two-premise deduction cites s_1, s_2. Their scopes are disjoint, so revision may combine their evidence if the reasoning system licenses revision for disjoint evidence.*

Counterexample 2.4 (Payload difference is not independence)**.** *Two packets may use different syntax, different truth values, and different derivation trees while both cite s_0. They are not source-independent. Changing representation cannot hide shared provenance.*

Counterexample 2.5 (Counts are insufficient)**.** *The scopes $\{s_0\}$ and $\{s_1\}$ both have cardinality one and are disjoint; $\{s_0\}$ and $\{s_0\}$ also both have cardinality one and overlap completely. No test using only source counts can distinguish the cases.*

This substrate belongs to neither NARS nor PLN. Both theories may import it; neither is defined as an instance of the other. Any operation stronger than scope inspection remains system-specific.

3 Two languages: beliefs are not their evidence

Wang separates a belief language L_B from an evidence language L_E and studies a degree map

$$d : L_B \times L_E \longrightarrow V,$$

where V is the chosen value space [15]. This separation prevents a recurrent type error: treating a proposition and the body of evidence used to assess it as if they were the same object.

An abstract two-language layer may be represented by:

- a type B of beliefs;
- a type E of evidence states;
- a value type V;
- a degree function $d : B \to E \to V$;
- a source-scope map on evidence;
- optional transformations of belief and evidence languages, with naturality laws for d.

Two equivalences must then be kept distinct:

$$b_1 \equiv_{\text{sem}} b_2 \qquad \text{and} \qquad e_1 \equiv_{\text{ev}} e_2.$$

Semantic equivalence of beliefs does not imply interchangeability of evidence bodies. Conversely, a shared evidence packet may support distinct claims.

3.1 Finite views are projections, not foundations

Wang objects to a general-purpose theory whose hypothesis and evidence spaces are fixed finite sets in advance [15]. Finite counting is useful, but it should be exposed as a finite view of a more general layer:

$$\text{open reasoning layer} \longrightarrow \text{finite observable fragment.}$$

The projection may support exact cardinal calculations without implying that the world, language, or future problem class is closed.

Example 3.1 (Hempel-style loss witness)**.** *The claims "raven implies black" and "non-black implies non-raven" may be classically contraposed at the belief level. Their evidence scopes need not coincide: observations collected under the first description need not be the observations collected under the second. A one-language quotient can erase this difference even when the propositions agree extensionally.*

4 NARS: truth values are path-relevant

NARS is designed for the Assumption of Insufficient Knowledge and Resources (AIKR). Under AIKR, all relevant evidence cannot be known or globally reconciled at every moment. Wang therefore permits the same statement to carry different truth values reached along different inference paths [15].

Represent a path-scoped belief as

$$\beta = (\varphi, \tau, \pi),$$

where φ is a statement, τ a NARS truth value, and π an evidence path retaining both rule occurrences and source identities. Its scope is derived from the path:

$$\mathsf{scope}(\beta) = \mathsf{scope}(\pi).$$

Theorem 4.1 (Single-event obstruction)**.** *Suppose two path-scoped beliefs have the same statement φ but different frequencies $f_1 \neq f_2$. No model assigning one measurable event E_φ to φ can realize both frequencies as $\Pr(E_\varphi) = f_i$.*

The proof is immediate: one event under one probability measure has one probability. The theorem is deliberately narrow. It does not claim that an individual NARS truth value lacks a probabilistic reading. It shows that the path-indexed family cannot be collapsed to one globally coherent event value without losing information.

4.1 Constructive incoherence

The resulting incoherence is not arbitrary contradiction. It is structured by retained evidence paths:

- disjoint evidence scopes license revision by evidence aggregation;
- overlapping scopes prohibit naive addition, avoiding double-counting;
- a choice policy may select one competing belief when revision is not licensed;
- later information may trigger a different reconciliation.

Thus path relevance provides a disciplined form of inconsistency tolerance. It is related to paraconsistency but not identical to it. A paraconsistent logic specifies which conclusions follow in the presence of conflict. Path-scoped evidence additionally records *how* each conflicting value was produced and whether their evidence may be combined.

Counterexample 4.2 (Why truth-value equality is too weak)**.** *Two beliefs may carry the same numerical truth value while one derives from $\{s_0\}$ and the other from $\{s_1, s_2\}$. Equality of truth values does not license evidence deduplication, revision, or replacement.*

5 PLN and constructible duality

PLN often represents evidence with separate positive and negative support. At a minimal level, an evidence object may be written

$$e = (e^+, e^-),$$

and classified by the constructive availability of positive and negative witnesses. This yields four qualitative regions:

	e^- absent	e^- present
e^+ absent	neither	negative only
e^+ present	positive only	both

The "both" region is not explosion; it is a state containing evidence on both sides. Source-scoped path identity refines this four-way classification. Two packets can both lie in the "both" region while citing different sources, carrying different multiplicities, or supporting different revision rights.

Principle 5.1 (Sibling theories over shared evidence geometry)**.** *NARS and PLN should share finite source geometry and generic proof-relevant relations. Their truth-value calculi, inference rules, revision policies, and probability interpretations remain separate. Correspondences are expressed as bridge theorems, not directory nesting or definitional equality.*

This organization prevents two opposite errors:

1. reducing NARS to a special case of PLN because both use evidence;
2. duplicating source-scope and independence theory inside each system.

5.1 The Wang–Walley numerical seam

Wang observes that the NARS frequency interval

$$\left[\frac{w^+}{w+k}, \frac{w^+ + k}{w+k}\right]$$

has the same numerical form as a Walley-style binary predictive interval

$$\left[\frac{t^+}{t+s_0}, \frac{t^+ + s_0}{t+s_0}\right]$$

after identifying $w^+ = t^+$ and $k = s_0$ [15, 14]. The identity is algebraic, not semantic. Walley's lower-prevision theory imposes coherence conditions; NARS deliberately does not require one globally coherent probability model over all path-relative beliefs.

Counterexample 5.2 (Equal interval, unequal evidence)**.** *Two source scopes of equal cardinality can produce the same interval while overlapping differently. The interval cannot determine whether revision is licensed. Therefore the interval does not factor the full evidence semantics.*

6 A common abstraction for razors

Ockham, MDL, MML, Bayesian evidence, Solomonoff induction, structural risk, Bennett weakness, and quantale weakness are not instances of one canonical scalar called "simplicity." Their most permissive common structure is weaker.

Definition 6.1 (Constrained selection criterion)**.** *For candidates C, a criterion consists of:*

(i) an admissibility predicate $A : C \to \mathsf{Prop}$;

(ii) a preference preorder $\succeq \subseteq C \times C$.

A candidate is optimal when it is admissible and is at least as preferred as every admissible competitor. It is maximal when no admissible competitor strictly dominates it.

Only reflexivity and transitivity are required. This preserves ties and incomparability. A scalar score, total order, probability interpretation, or quantale multiplication is additional structure supplied by a particular profile.

Profile	Admissibility	Preference
Ontological parsimony	satisfies necessity conditions	fewer posited entities
MDL/MML	valid model/code	shorter model-plus-data message
Bayesian evidence	model class accepted	larger marginal likelihood
Solomonoff	representation fixed	larger algorithmic semimeasure
Structural risk	certified learning setup	smaller risk plus capacity penalty
Bennett	correct policy for the task	more semantic completions
Quantale weakness	relation admitted	larger quantale-valued relational score

6.1 Is Bennett's Razor the master abstraction?

No. Bennett's Razor is a particularly important semantic profile: among correct policies, prefer the policy compatible with the largest set of completions [2, 3]. It captures least commitment in the model space. But it does not subsume criteria whose primitive data are description length, marginal likelihood, computational cost, or a multi-dimensional tradeoff.

Counterexample 6.2 (Bennett–description disagreement)**.** *Let candidate h_w have two semantic completions and a two-bit code. Let h_s have one completion and a one-bit code. Bennett strictly prefers h_w; description length strictly prefers h_s. Under the product criterion the two candidates are incomparable. Neither razor globally subsumes the other.*

Goertzel's quantale profile is more compositional than a cardinal count: a complete lattice represents partial comparison, and monoidal multiplication combines relational evidence [4]. Yet it too is a profile above the neutral criterion, not the definition of every selection problem.

6.2 Applying weakness to formalization itself

Let a formalization candidate F have a model class $\mathsf{Models}(F)$ and let $\mathcal{O}$ be an explicit set of adequacy obligations. Define

$$F_1 \succeq_{\mathrm{sem}} F_2 \quad \Longleftrightarrow \quad \mathsf{Models}(F_2) \subseteq \mathsf{Models}(F_1),$$

but compare only candidates satisfying every obligation in $\mathcal{O}$. The optimal formalization is then the least committing adequate one.

This is not an excuse for vacuity. The unconstrained theory that admits every model fails any obligation requiring false models to be excluded. The inconsistent theory that admits no models may satisfy negative conditions vacuously, but loses to an adequate theory with a larger nonempty model class.

Principle 6.3 (Reflexive weakness discipline)**.** *Choose the weakest interface that proves the required positive and negative theorems. Add scalar scores, total decisions, finite assumptions, or merge policies only in named profiles. Recover each lossy view by theorem.*

The neutral source-scoped interface is an application of this discipline: it shares exactly the provenance structure needed by NARS and PLN while refusing to impose either system's truth semantics on the other.

7 When weakness predicts generalization

Bennett's finite generalization result considers a uniform distribution over subsets of unseen demands. If a candidate handles m elements of an unseen domain of size n, then its success probability is

$$\frac{2^m}{2^n}.$$

On this explicit uniform sample space, weakness and success probability induce the same order [2].

The assumption is mathematically load-bearing. Özkural studies task sequences with nonuniform structure and distinguishes four generative models [9]:

1. independent uniform fixed-width programs (random typing);
2. independent zeta-ranked programs;
3. programs assembled from a shared subprogram database;
4. evolutionary zeta mutations, retaining the old program after an invalid mutation.

All four fit a history-indexed task process

$$T : \mathsf{List}(X) \longrightarrow \mathsf{PMF}(X).$$

For a handling relation $H(c, x)$, define the next-task success probability $p_T(c \mid h)$. The exact question is not whether the process has an interesting name, but whether it preserves the weakness ordering at history h:

$$p_T(c_1 \mid h) \le p_T(c_2 \mid h) \iff w(c_1) \le w(c_2).$$

Proposition 7.1 (Sufficient factorization certificate). *If there exists a strictly increasing curve $g : \mathbb{N} \to [0, 1]$ such that*

$$p_T(c \mid h) = g(w(c))$$

for every candidate c, then the process preserves the weakness order at h.

Counterexample 7.2 (Biased future law). *Two candidates can have equal completion counts while covering differently weighted tasks. Under a biased task law they have different success probabilities. Cardinal weakness therefore cannot universally rank generalization under arbitrary nonuniform processes.*

The productive synthesis is conditional: Bennett supplies the semantic freedom order; Özkural supplies explicit families of task laws; a theorem or certificate must state when the law preserves the order. Distributional structure is not silently converted into a universal transfer theorem.

8 Three priors that should remain distinct

Three programme-level priors illustrate why the common abstraction should not be a single formula.

Solomonoff. Algorithmic probability sums the weights of programs producing an output, relative to a prefix machine [11]. It is representation invariant only up to the usual machine-dependent constants and is generally not computable.

Bennett. For a finite implementable language L_v, Bennett's task-coverage weight is proportional to the number of subsets covered by a hypothesis. Normalization over the finite hypothesis language preserves the weakness ordering [2].

Özkural. For exponent $s > 1$, the zeta rank law

$$P(k) = \frac{1}{k^s \zeta(s)}, \qquad k \geq 1,$$

is a normalized, computable rank distribution. Transport to programs is automatically normalized when the representation supplies a duplicate-free enumeration. Binary strings with leading zeros are a negative control: naive binary arithmetization is not injective and cannot justify normalization by bijection [9].

Concrete hypotheses can be chosen so that the zeta rank, Solomonoff-shaped program-length weight, and Bennett semantic-freedom weight each prefer a different hypothesis. This ordering disagreement is evidence of conceptual independence, not a defect to scalarize away.

9 Architectural synthesis without attribution collapse

Özkural's Omega is an architectural synthesis rather than a renamed Solomonoff or Gödel machine. Its source-faithful decomposition separates [8]:

- the inherited Alpha substrate and algorithmic-probability programme;
- proof-backed reflective self-improvement inherited from the Gödel machine programme;
- an explicit AI-kernel bootstrap premise;
- open-ended problem-solving-method invention and retention;
- Heuristic Algorithmic Memory over multiple reference machines;
- analysis and synthesis of methods, forecast-guided ensembles, and operation without assuming a substantially known environment.

The bootstrap remains a premise with evidence, not a theorem manufactured by the architecture record. Likewise, an inherited universal-machine component should reuse the underlying Solomonoff construction rather than reattribute it to Omega. Precise decomposition honors both lineage and novelty.

Omega and OmegaClaw are distinct architectures. Similar names and overlapping topics do not justify identifying their components. Until the primary OmegaClaw text is available, metadata cannot supply its definitions or correctness obligations.

10 Bounded control and observation aliasing

Hammer and Lofthouse describe OpenNARS for Applications (ONA) as a bounded architecture with approximately constant-time cycles, priority-guided attention, and explicit forgetting [6]. Relative forgetting changes the ranking used for attention; absolute forgetting evicts items to enforce a memory bound. A typical normalized usefulness map has the form

$$u \longmapsto \frac{u}{u+1},$$

which is monotone and lies in $[0, 1)$ for nonnegative u.

These control laws belong above NARS evidence semantics. They govern which items receive scarce computation; they do not redefine the truth functions. This separation permits other bounded controllers to operate over the same evidence calculus.

10.1 History can carry decision-relevant information

Hammer's ONA comparison includes tasks where the current observation aliases states requiring different actions [5]. Let h_0, h_1 be two histories and r a readout retaining only the current observation. Suppose

$$r(h_0) = r(h_1), \qquad a_0 \neq a_1,$$

and correctness requires action a_i at h_i. Then no deterministic policy π depending only on $r(h)$ can satisfy both requirements, because it must assign one action to their shared readout.

Theorem 10.1 (Observation-fibre obstruction)**.** *If one observation fibre contains two histories with incompatible admissible decision sets, no policy factoring through that observation readout realizes the full history-sensitive specification.*

Example 10.2 (Positive history policy)**.** *A policy retaining the last action can continue the previous motion in both histories even when the visible observation is identical.*

Counterexample 10.3 (The obstruction is decision-relative)**.** *A constant policy still factors through the lossy current-observation readout. Therefore partial observation is not always harmful; it is harmful exactly when it merges histories whose required decisions differ.*

Hammer's reported experiments and the later independent comparison by Beikmohammadi and Magnússon should be represented as source-scoped empirical records, not promoted to a theorem that ONA universally dominates Q-learning [5, 1]. The theorem concerns observation factorization. Performance is a separate empirical question with protocol, encoding, and abstention choices retained in its receipt.

11 Architectural synthesis

The dependency structure is intentionally not a hierarchy with PLN at the root or NARS at the root:

$$\begin{tikzcd}
& \text{finite source geometry and proof-relevant relations} \arrow[d] & \\
& \text{source-scoped evidence interface} \arrow[dl] \arrow[dr] & \\
\text{NARS core} \arrow[d] & & \text{PLN core} \arrow[d] \\
\text{NARS control} & & \text{PLN rule families}
\end{tikzcd}$$

The two-language layer, enactive task theory, Universal AI priors, GSLT observations, and cognitive architectures are additional clients. Cross-links live in explicit bridge modules with theorem statements naming exactly what is preserved and what is forgotten.

Layer	Retains	Must not assume
Source geometry	occurrence identity, finite scope, overlap	truth-value or merge semantics
NARS evidence	paths, scopes, NARS truth values	one global probability event per statement
PLN evidence	positive/negative support, rule evidence	NARS revision or control policy
Razor profile	admissibility and preference preorder	total scalar comparison
Task process	history-indexed next-task law	weakness preservation without proof
Control layer	resource allocation and forgetting	change to underlying truth semantics
Observation layer	witness container and readout	reflection from lossy value to history

12 Implications for a proof-relevant cognitive language

A MeTTa-like language performs nondeterministic pattern matching and may retain multiple derivations of one answer. Its natural semantic relation is proof-relevant:

$$\mathsf{Rel}(A, B) = A \to B \to \mathsf{Type}.$$

Relational composition should retain the intermediate witness and both derivations:

$$\mathsf{Chain}(R, S)(a, c) = \sum_{b:B} R(a, b) \times S(b, c).$$

The existential witness b is not administrative bookkeeping. It identifies the intermediate result, supports source receipts, and contributes to answer multiplicity. A truth relation or Boolean success flag is the support quotient of this richer object.

Source-scoped evidence then attaches provenance to derivations; NARS- or PLN-specific readouts may observe those derivations without replacing them. Likewise, a decision interface should distinguish

$$\mathsf{Established}(e), \quad \mathsf{Refuted}(o), \quad \mathsf{OutsideFragment}(r), \quad \mathsf{Incomplete}(b, f),$$

with infrastructure faults outside the semantic sum. Failure to find a proof is not a checked obstruction, and lack of fragment coverage is not falsehood.

This arrangement supports cognitive self-description without confusing the theory with its checker. The mathematical language may internalize evidence, outcomes, and receipts at a higher universe or stage; an executable checker is one consumer of that theory. Truth readouts, authority decisions, and cost observations remain orthogonal projections unless a theorem connects them.

13 Formalization map

The accompanying Lean development is organized by mathematical ownership:

- `Mettapedia.Evidence.SourceScope`: finite source geometry;
- `Mettapedia.Evidence.SourceScoped`: the heterogeneous packet interface over that geometry;
- `Mettapedia.Evidence.TwoLanguage`: Wang's two-language evidence interface;
- `Mettapedia.NARS`: NARS truth functions, inference, evidence paths, and bounded control;
- `Mettapedia.NARS.Bridges`: explicit PLN, probability, cognitive-architecture, and Universal-AI comparisons;
- `Mettapedia.PLN`: PLN's own evidence and rule theories;
- `Mettapedia.Enactive`: Bennett's finiteness-free and finite layers, razor profiles, and task-process generalization;
- `Mettapedia.UniversalAI`: Solomonoff, weakness, zeta, and self-improving-agent constructions;
- `Mettapedia.GSLT`: proof-relevant dynamics, observation disciplines, and transport laws.

Each main concept has positive and negative controls. The negative controls are part of the specification: they distinguish source identity from source count, numerical equality from semantic identity, admissibility from vacuity, and process structure from a generalization guarantee.

14 Open obligations and source boundaries

Open obligation 14.1 (Infinite and open evidence). *Finite scopes are suitable receipts for finite computations, but the reasoning theory should permit open vocabularies and growing evidence languages. Finite views must be proved conservative projections of the open layer where used.*

Open obligation 14.2 (Credal and path-scoped integration). *Develop lower and upper expectations over source-scoped path families without imposing global coherence on NARS. The target should say which local families are coherent and how incompatible scopes coexist.*

Open obligation 14.3 (Task-process transfer theorems). *For each Özkural process model, identify sufficient and necessary conditions under which success probability factors through weakness. Do not infer the condition from a zeta label or from power-law motivation alone.*

Open obligation 14.4 (Bounded control adequacy). *Relate ONA's attention and forgetting mechanisms to a formal anytime policy, including memory invariants, fairness/starvation properties, and source-scoped empirical conformance.*

Open obligation 14.5 (Primary-source gate). *Architectural claims from conference papers whose full texts are unavailable must remain unformalized. Bibliographic metadata and abstracts can establish that a work exists, but not its definitions, theorem hypotheses, or algorithms.*

15 Conclusion

Path relevance is the missing coordinate behind many apparent conflicts among truth-value systems. A statement's scalar value does not say which path produced it, which sources it consumed, whether another value is independent, or whether the two may be revised together. Retaining those distinctions makes NARS's local incoherence mathematically intelligible, refines PLN's constructive positive/negative geometry, and gives bounded control systems an honest provenance substrate.

Bennett's Razor then occupies its proper place: a powerful semantic least-commitment profile, not the universal definition of simplicity. Goertzel's quantale weakness adds compositional relational structure; Özkural's stochastic task laws identify the distributional boundary of weakness-based generalization; Hammer's work connects the theory to bounded continual control. Their common abstraction is deliberately modest: admissible candidates ordered by a preorder, evidence packets exposing source scope, and explicit theorems for every stronger identification.

This is also a discipline for formalization itself. Preserve the most permissive adequate model class; place system-specific commitments in named profiles; retain evidence before taking quotients; and make every loss of information visible at a theorem boundary.

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